

Martin Lavin Carvajal Portfolio

Personal portfolio for **Martin Lavin Carvajal**, Computer Engineering student focused on cloud engineering, AWS architecture, backend systems, and AI-powered applications.

The site presents selected projects as professional case studies, with optional live demos, architecture diagrams, and repository links.

Stack

Layer	Technology
Framework	Next.js 16 (App Router)
Language	TypeScript
UI	React 19, Tailwind CSS 4
Motion	Framer Motion
Fonts	Geist

Features

- Dark, minimalist Vercel-inspired layout
- Hero, Projects, About, Skills, Contact, and Footer sections
- Project cards with thumbnails, tech tags, GitHub, Case Study, and Live Demo (when available)
- Dynamic case study pages at `/projects/[slug]`
- Embedded live demos for selected projects (iframe with fallback)
- Architecture diagrams and structured case study sections
- Subtle site-wide starfield background with reduced-motion support

- CV downloads (English / Spanish) from the Hero section

Featured projects

Project	Focus
Document Knowledge Agent	Cloud-native RAG on AWS
Cloud Operations Lab	Terraform, CI/CD, OIDC, SSM, observability
ECG-AI Serverless	ML inference on AWS Lambda
Skill Tracker	Full-stack Go + React + PostgreSQL

Project structure

app/	# Next.js App Router pages and layout
components/	
layout/	# Navbar, Footer, cosmic background
sections/	# Hero, Projects, About, Skills, Contact
projects/	# Case study UI building blocks
ui/	# ProjectCard, LiveDemoEmbed
motion/	# Page and reveal animations
data/	# Projects, case studies, about, skills
case-studies/	# Per-project case study content
lib/	# Helpers (projects, motion)
public/	# Static assets (project images, CV)
types/	# Shared TypeScript types

Content is separated from UI: update `data/projects.ts` and `data/case-studies/*` to change portfolio copy without touching components.

Getting started

Requirements

- Node.js 20+ (recommended)
- npm

Install

```
npm install
```

Development

```
npm run dev
```

Open <http://localhost:3000>.

Build

```
npm run build  
npm start
```

Lint

```
npm run lint
```

Scripts

Command	Description
<code>npm run dev</code>	Start the development server
<code>npm run build</code>	Create a production build
<code>npm start</code>	Serve the production build
<code>npm run lint</code>	Run ESLint

Adding or updating a project

1. Add or update the entry in `data/projects.ts`.
2. Add or update the case study in `data/case-studies/<slug>.ts`.
3. Place images under `public/projects/` (thumbnail, architecture diagrams, etc.).
4. Set optional fields as needed:
 - `liveUrl` for Live Demo + embedded demo
 - `architectureImage` for the Architecture section
 - `githubUrl` for the repository button

Author

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